



Tuberculosis Diagnosis

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INTRODUCTION

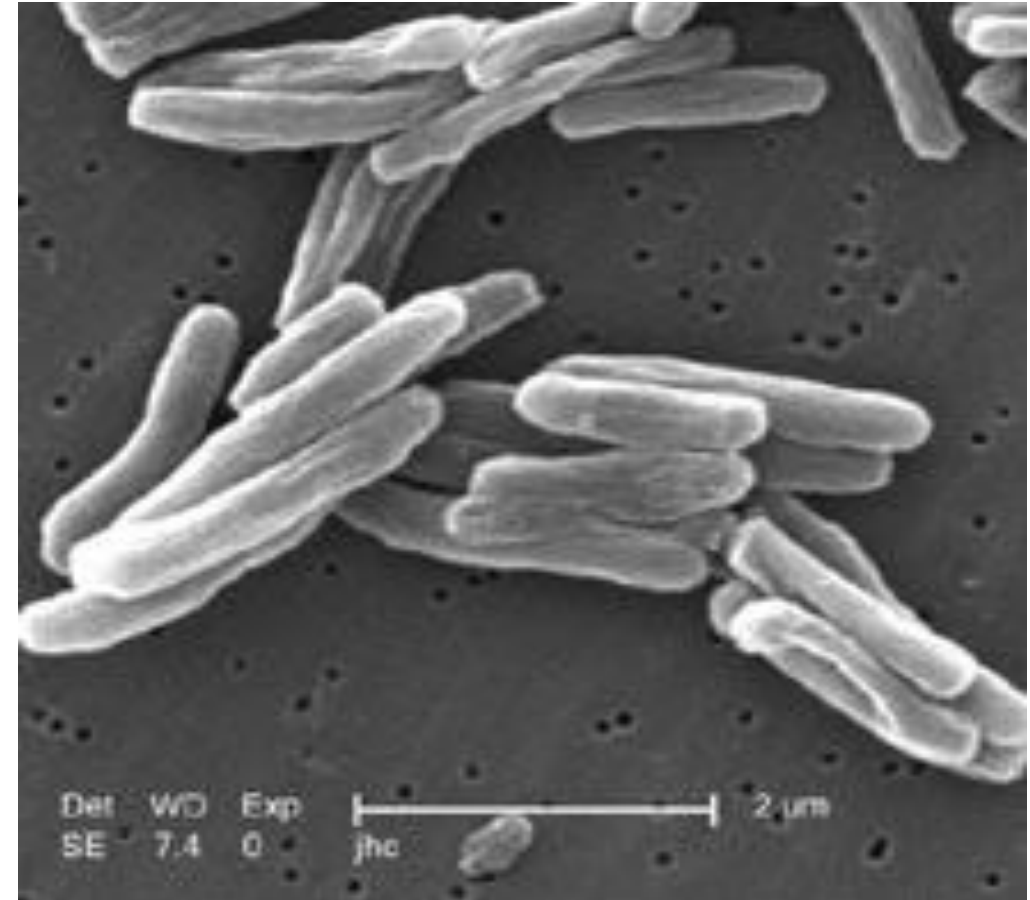
TUBERCULOSIS:

TB is an infectious disease caused mainly by the bacillus *Mycobacterium tuberculosis*, a small, aerobic, nonmotile gram positive bacillus.

MODE OF SPREAD:

Infectious aerosol droplets 0.5 to 5.0 µm in diameter.
A single sneeze can release up to 40,000 droplets.

Each one of these may transmit the disease, since the infectious dose is very small (the inhalation of **fewer than 10 bacteria** may cause an infection).



GLOBAL BURDEN OF DISEASE

- There were an estimated 10.7 million new cases of TB in 2024; 15% children, 6% PLHIV
- 1.1 million died from the disease (150,000 among HIV positive people)
- An estimated 450,000 people developed MDR- TB,

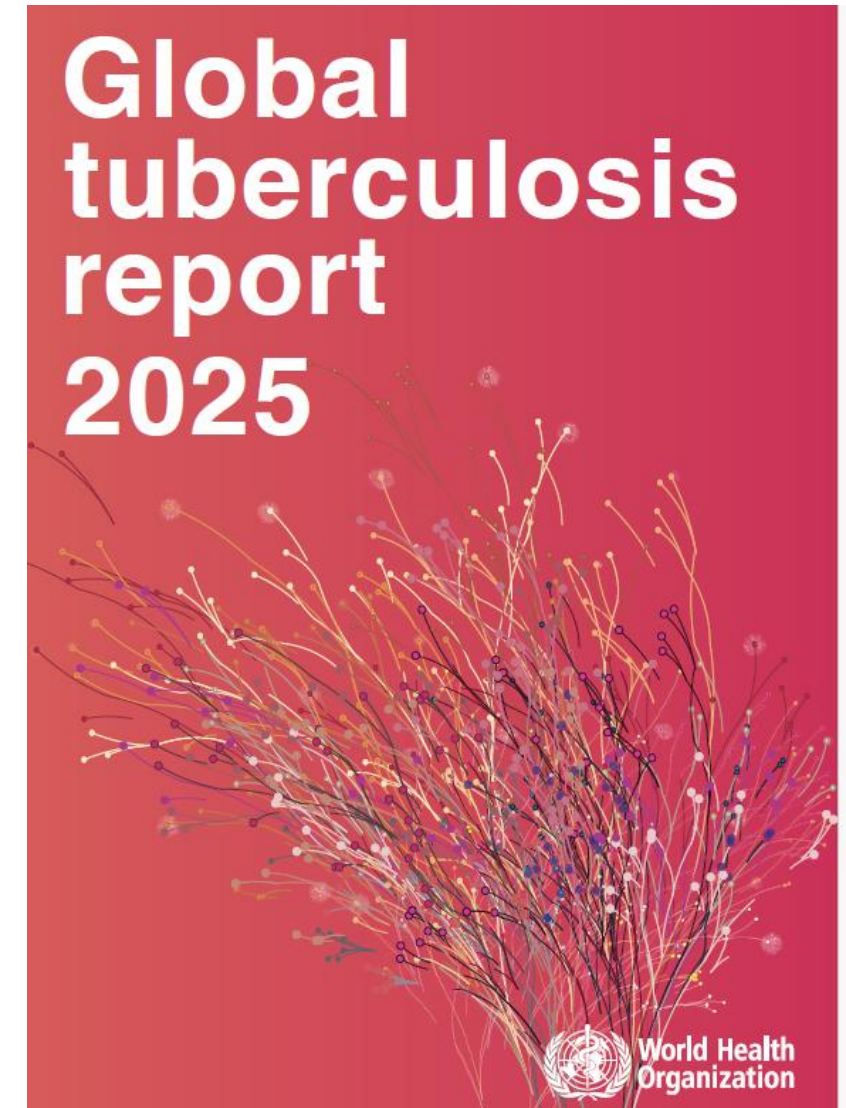


TABLE 3.1

Targets for percentage reductions in TB disease burden set in WHO’s End TB Strategy

INDICATORS	MILESTONES		TARGETS	
	2020	2025	2030	2035
Percentage reduction in the absolute number of TB deaths (<i>compared with 2015 baseline</i>)	35	75	90	95
Percentage reduction in the TB incidence rate (<i>compared with 2015 baseline</i>)	20	50	80	90 (~10/100 000 population)

END TB STRATEGY



A. Early diagnosis of TB including universal drug-susceptibility testing, and systematic screening of contacts and high-risk groups

B. Treatment of all people with TB including drug-resistant TB, and patient support



D. Preventive treatment of persons at high risk; and vaccination against TB

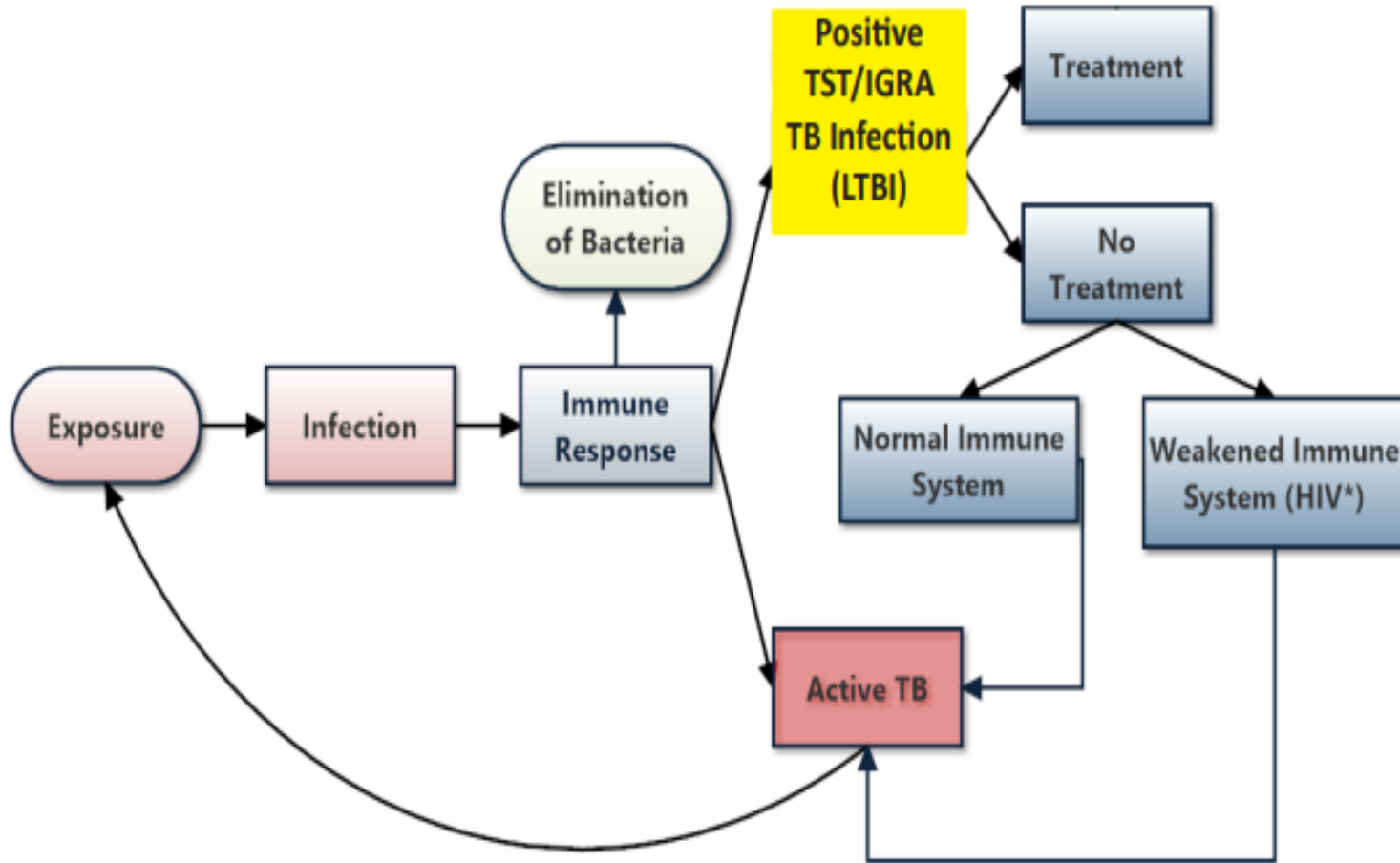
C. Collaborative TB/HIV activities; and management of co-morbidities



Why focus on Diagnostics

- Limited access to adequate initial diagnosis.
- About 3–4million people (25–35 % of the annual estimated incidence) likely undiagnosed.
- About 10 -15 % are children under 5years
- 25% of adults and 80% of children managed on the basis of clinical suspicion.
- Access to diagnosis is particularly challenging in children with TB.
- Faster, accurate TB diagnosis saves lives, cuts transmission, detects drug resistance, and closes the large “missing” case gap.

TB disease spectrum



- ❖ Pulmonary TB
- ❖ TB Adenitis
- ❖ Abdominal TB
- ❖ TB Pericarditis
- ❖ TB Osteomyelitis
- ❖ TB Meningitis
- ❖ TB (any other) body organ

Specimen for TB



Respiratory
specimen



Stool



Cerebral Spinal
Fluid



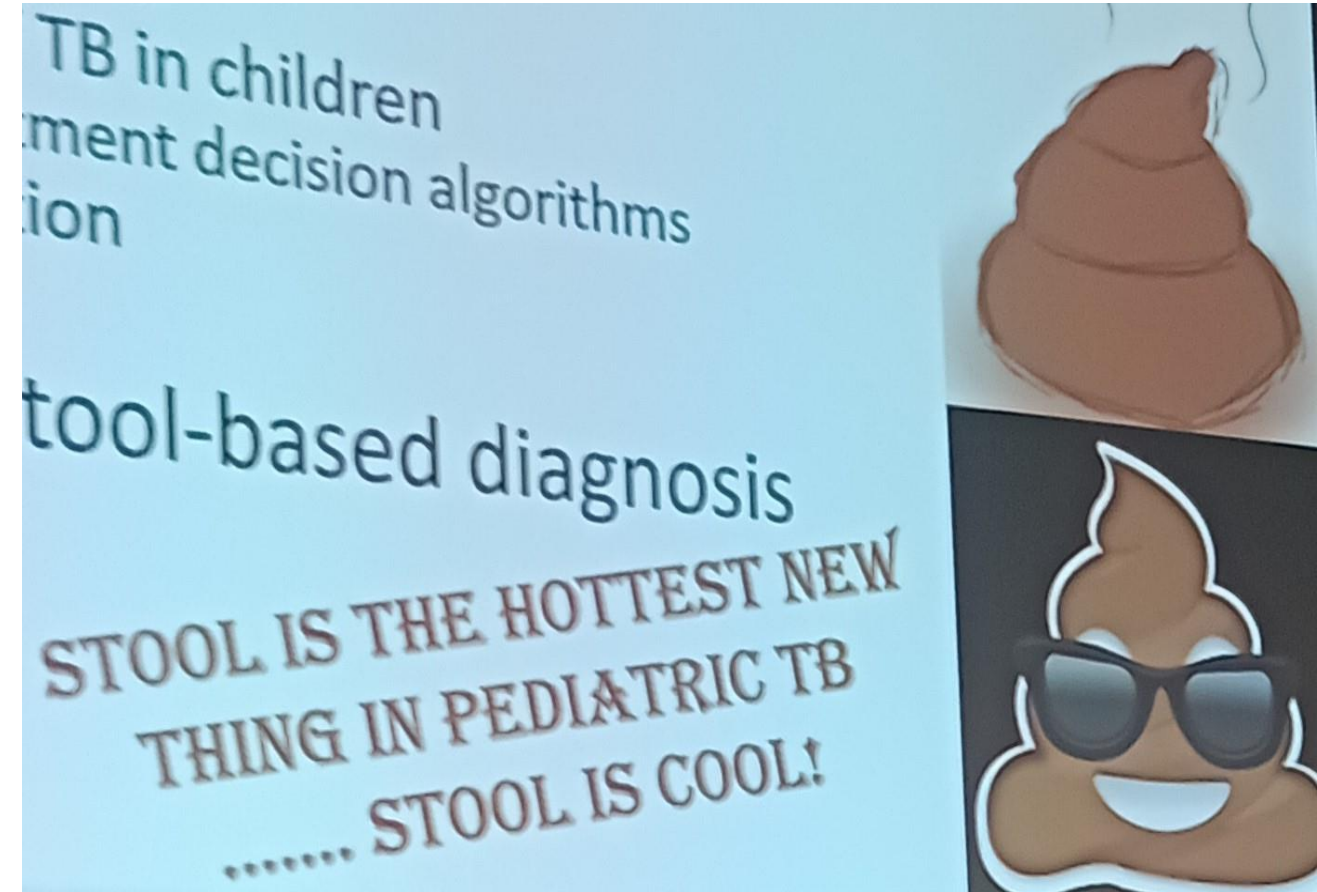
Plural Fluid



Histology



Oral Swabs



TB Diagnostics

Direct

- Microscopy
- Culture-based Diagnostics
- Molecular detection
- Breath biomarker – detection

Indirect

- Clinical Signs and Symptom
- Cellular response-
- Radiology
- Virbo Acoustic Signatures

Validity of a Diagnostic Test

- Sensitivity = $P(\text{Test+} \mid \text{Disease+})$
- Specificity = $P(\text{Test-} \mid \text{Disease-})$

○ Sensitivity =

$$\frac{\text{number of true positives}}{\text{number of true positives} + \text{number of false negatives}}$$

○ Specificity =

$$\frac{\text{number of true negatives}}{\text{number of true negatives} + \text{number of false positives}}$$

SMEAR MICROSCOPY

METHODS: (commonly used)

- Zeihl neelsen technique (hot acid fast stain)
- Kinyoun technique (cold acid fast stain)
- Fluorescent technique (10% sensitivity vs ZN, $\geq 46\%$ less time)
- Vital staining technique (differentiate between live and dead bacilli)
- **Automated microscopy**



Sensitivity 47-60%, Specificity 95-98%

Mycobacterial culture

- Gold standard for diagnosis
- Can detect as few as 10 viable bacilli per millimeter of sample.
- Distinguishes even between individual members of MTBC by biochemical and phenotypic testing.
- Permits DST.
- Can take a 2 to 8 weeks.
- Requires level 3 biosafety measures.
- **Lowenstein –Jensen medium**

CULTURE MEDIA
SOLID MEDIA

▪ <u>Egg-based Media:</u>	▪ <u>Blood containing Media:</u>
▪ Lowenstein-Jensen (LJ) Medium	▪ Tarshi's Medium
▪ Dorset Medium	
	▪ <u>Agar-based Media:</u>
▪ <u>Serum containing Media:</u>	▪ Middlebrook 7H10
▪ Loeffler's Medium	▪ Middlebrook 7H11
▪ <u>Potato-based Media:</u>	▪ Middlebrook Biplane (7H10/7H11 S Agar).
▪ Pawlowsky's Medium	

BACTEC TB-460

- semi-automated system
- Medium –modified middle brook 7H9 medium
- Palmitic acid is radiolabeled with C¹⁴

(MGIT) Mycobacterial Growth Indicator Tube

- Consists of round bottom tubes containing 4 ml of modified Middlebrooks 7H9 broth.
- Oxygen sensitive fluorescent sensor.
- Depletion of the dissolved oxygen in the broth & allow the indicator to fluoresce brightly

CULTURE MEDIA LIQUID MEDIA

- BACTEC 12 B Medium
 - BACTEC 460 TB
 - BACTEC 9000 MB
 - BACTEC MGIT 960
 - EPS Culture System II
 - Middlebrook 7H9 Broth
 - SeptiChek AFB
 - Dubo's Medium
 - Tween 80 (Sorbitol Mono oleate)
- } Continuous Monitoring system

MOLECULAR METHODS

Polymerase Chain Reaction (PCR)

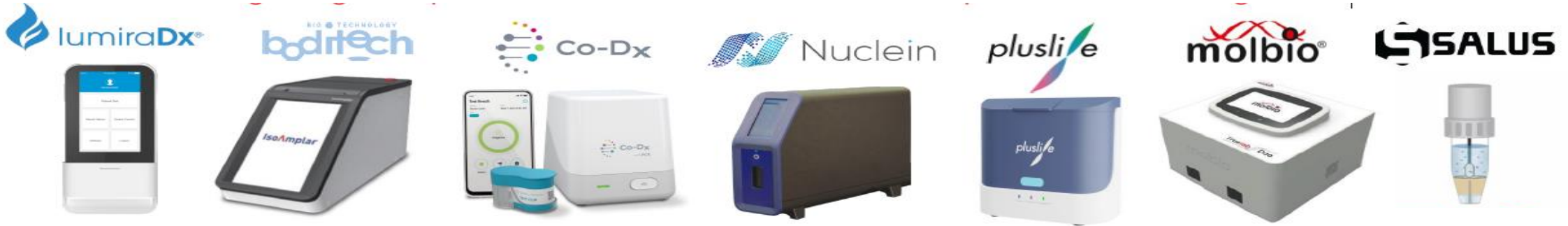
- Late 1980s–early 1990s: initial clinical PCR assays for *M. tuberculosis*,
- Mid-1990s: methodological improvements - better DNA extraction, contamination control) increased sensitivity/specificity.
- Late 1990s–early 2000s: adoption of real-time (quantitative) PCR platforms improved speed, quantification
- 2000s: line-probe assays (molecular assays for drug-resistance) and further optimized diagnostics.
- 2010: WHO endorsement of the fully automated Xpert MTB/RIF (real-time PCR with integrated sample prep)



PCR based Molecular Testing

PCR based Molecular tech	Sensitivity	Specificity	Features
Xpert MTB/RIF Ultra (Cepheid): 2010/2017	88–92%	(96–98%)	rifampicin resistance
Truenat MTB / MTB Plus (Molbio)- 2020	80–90	95–99	rifampicin resistance
Becton Dickinson (BD) MAX MDR-TB,	80-95%	95	rifampicin/INH markers
Roche cobas MTB (cobas MTB and cobas MTB/RIF)	75- 95%	95-99	rifampicin resistance
TB-LAMP (Eiken/other manufacturers; isothermal LAMP) 2016	60–70%	50–60%	No resistance detection
Line-probe assays (LPAs; e.g., Hain GenoType MTBDRplus	95–98%, INH resistance & 70–90% (depends on target mutations)	> 95%	

Newer Oral Swab based PCR Platforms



WHO recently approved near Point of Care(nPOCS) tests- These are Swab based.

- PCR based
- Less sophisticated , Cheaper
- Quicker turn around
- None tests for drug resistance

nPOC	Sensitiv	Specificity
Pluslife MiniDock- Oral Swab	76%	99%
Pluslife MiniDock – Dipped Sputum Swab	86%	98%
MTB –Ultima (Molbio	82-90%	99%

Stool testing

Molecular Testing

- Sensitivity 50 -65%, Lower in pauci bacillary disease
- Specificity 97-99%
- For use among children if sputum collection is not feasible
- Endorsed by WHO in 2023

Stool Microbiome:

- Quantifies commensals vs Opportunists:
- AUC 0.82-0.90

Step 1

Formed and semi formed (1-5)

0,8 to 1 gram (thumb nail)

Unformed and watery (6,7)

Replace 2 (6) or 4 (7) ml SR with stool

Step 2

Close bottle well, Shake 30 seconds, leave 10 min, shake again 30 sec and leave another 10 min, after last shake and sedimentation do NOT lift bottle

Step 3

Squeeze balloon well and carefully aspirate in one go 2 ml supernatant from upper layer, do NOT disturb sedimentations – do NOT lift bottle or touch sediment

XDR ASSAY

- The XDR assay is designed to be a reflexive test when an Xpert® (or Ultra) MTB/RIF positive test indicates an infection with RIF-resistant MTBC.
- The XDR assay will further genotype common resistance alleles to INH, FLQ, and AMG.
- Highly specific , but sensitivity varies by drug . 60- 95%, lowest for AMGs

Other molecular markers

RNA based markers

- Newer markers
- Nucleic Acid Sequence Amplification techniques
- Targets the TB 16s, 23s rRNA
- Growing importance in clinical monitoring
- Sensitivity 70-90% , 90-98% Specific

Cell Free DNA – cfDNA

- Evaluated in Urine and Plasma
- Uses Next Generation Sequencing technologies
- Sensitivity -30-60%
- Specificity 85-95%
- Still a pipeline marker

Other molecular markers

Proteomics

- Down regulated vs upregulated
- NG molecular tech – Dia PASEF , ML
- SFTPb, IGHV3-30, IGKV1-16 highly expressed
- APOM and KIT reduced
- Combined AUC -0.86-

Fossati, A., Wambi, P., Jaganath, D., Calderon, R., Castro, R., Mohapatra, A., ... & COMBO Study Consortium. (2025). **Plasma proteomics for biomarker discovery in childhood tuberculosis.** *Nature communications*, 16(1), 6657.

TB Metabolites.

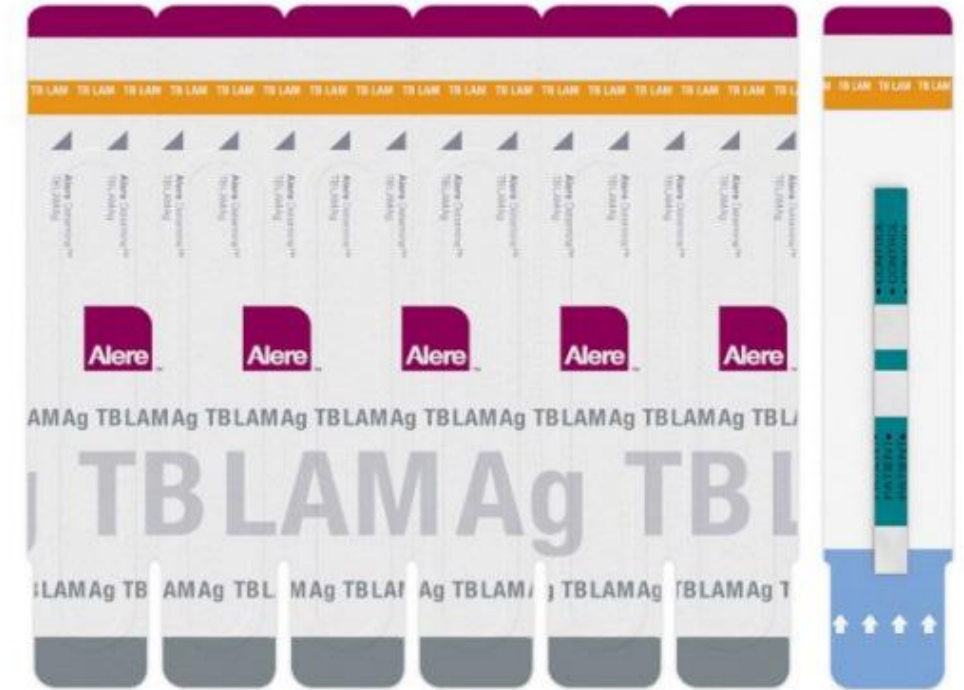
- Picolinate 0.66 ,
- Neopterin, 0.65
- Dihydroneopterin 0.66
- Combined signature AUC 0.72

Mousavian, Z., Segal, M. R., Calderon, R. I., Luiz, J., Nkereuwem, E., Wambi, P., ... & Jaganath, D. (2026). **A Multi-Omics Study Reveals Pathway-Level Insights and Predictive Biomarkers in Pediatric TB.** *medRxiv*, 2026-03.

Non – Sputum biomarkers to detect active tb

Determine TB LAM Ag rapid assay:

- It is an immuno chromatographic strip
- Targets the LAM Ag in urine via an antibody capture and detection.
- Not only specific for MTBc but also detects other NTMs.
- LAM is released in the blood and ultimately expelled by urine.
- Duration of test is 25 mins, minimal training required,
- Approved for use among adults with HIV



Sensitivity 25–45% / 50-70% , specificity 95-98%

The novel Breath tests

Assessing VOCs via simple breath tests for rapid screening of TB infection.

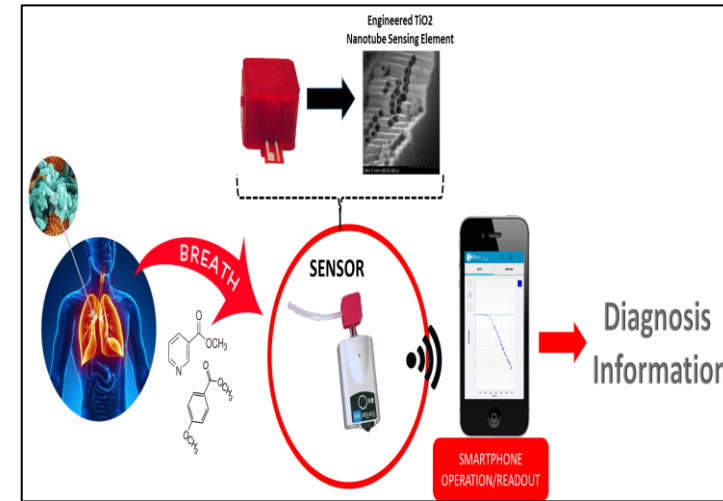
- Methyl P anisate
- Methy P Nicotinate

➤ Pipeline Diagnostics

➤ TB Breathalyzer

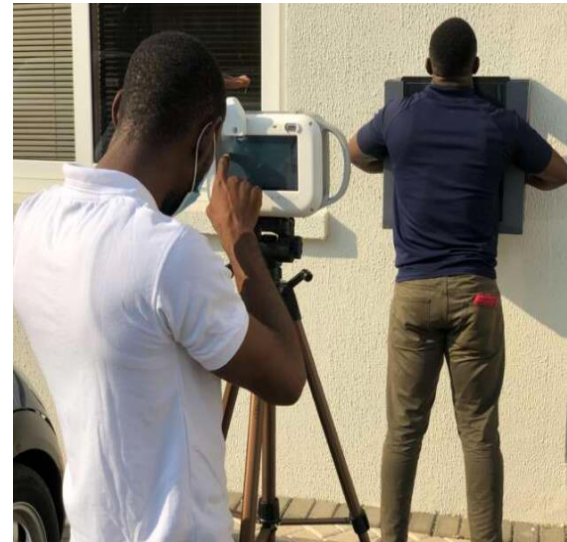
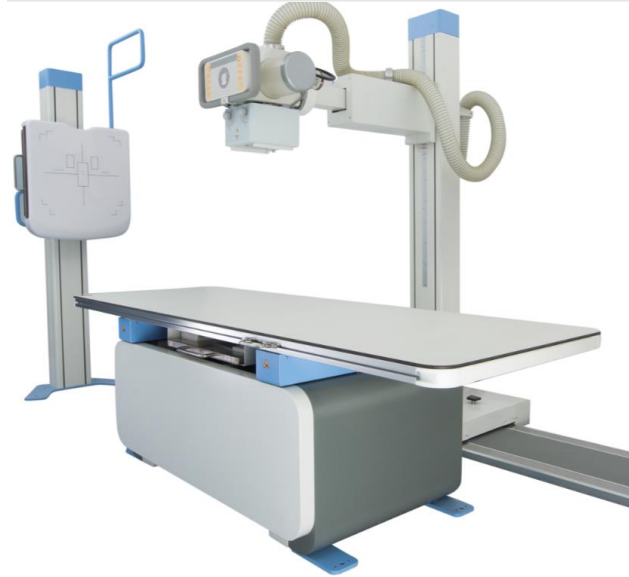
➤ TB Breath Sensor

- Sensitivity 70–95%,
- Specificity 60–90%.



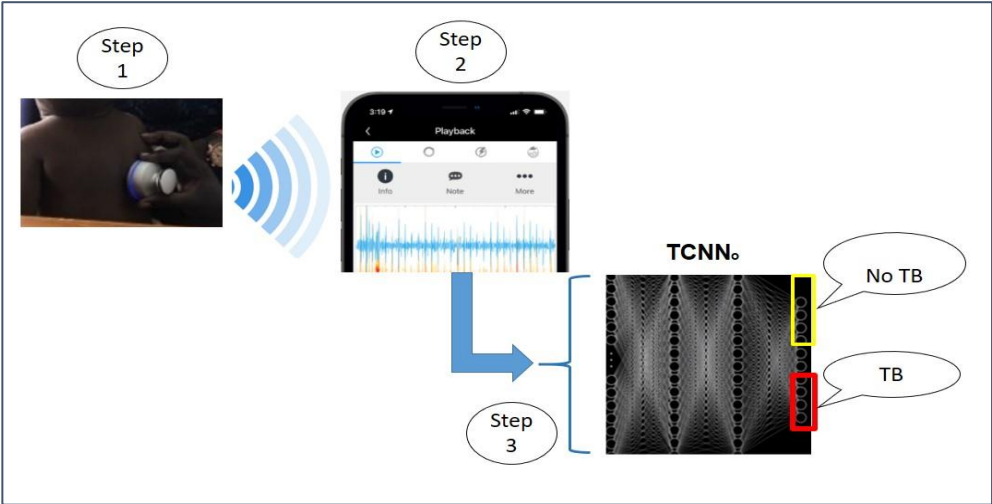
Chest XRAY

- CXR has potential use in both active and passive case finding of TB and can integrate with other diagnostic tests such as Xpert MTB\RIF for rapid identification of active TB.
- Used as a screening tool
- Fast
- low variable cost
- High CAD /AI potential
- Limited by the need for expertise to read the CXR.
 - Sensitivity 10-30%
 - Specificity 75-90%



AI- Supported candidate vibro acoustic signatures

Digital Signatures	Candidate Tool	Sensitivity / Specificiy
Automated Lung sound analysis	Stethee, GE, Ostium,	60 -77% / 80%
Automated cough sound analysis	Swaasa , Hyfe	90%/ 86%
Lung sound and Vibration analysis	Level 42ai- Impulse	AUC 0. 81



Diagnosis of Latent Tuberculosis

Tuberculin skin test(TST)

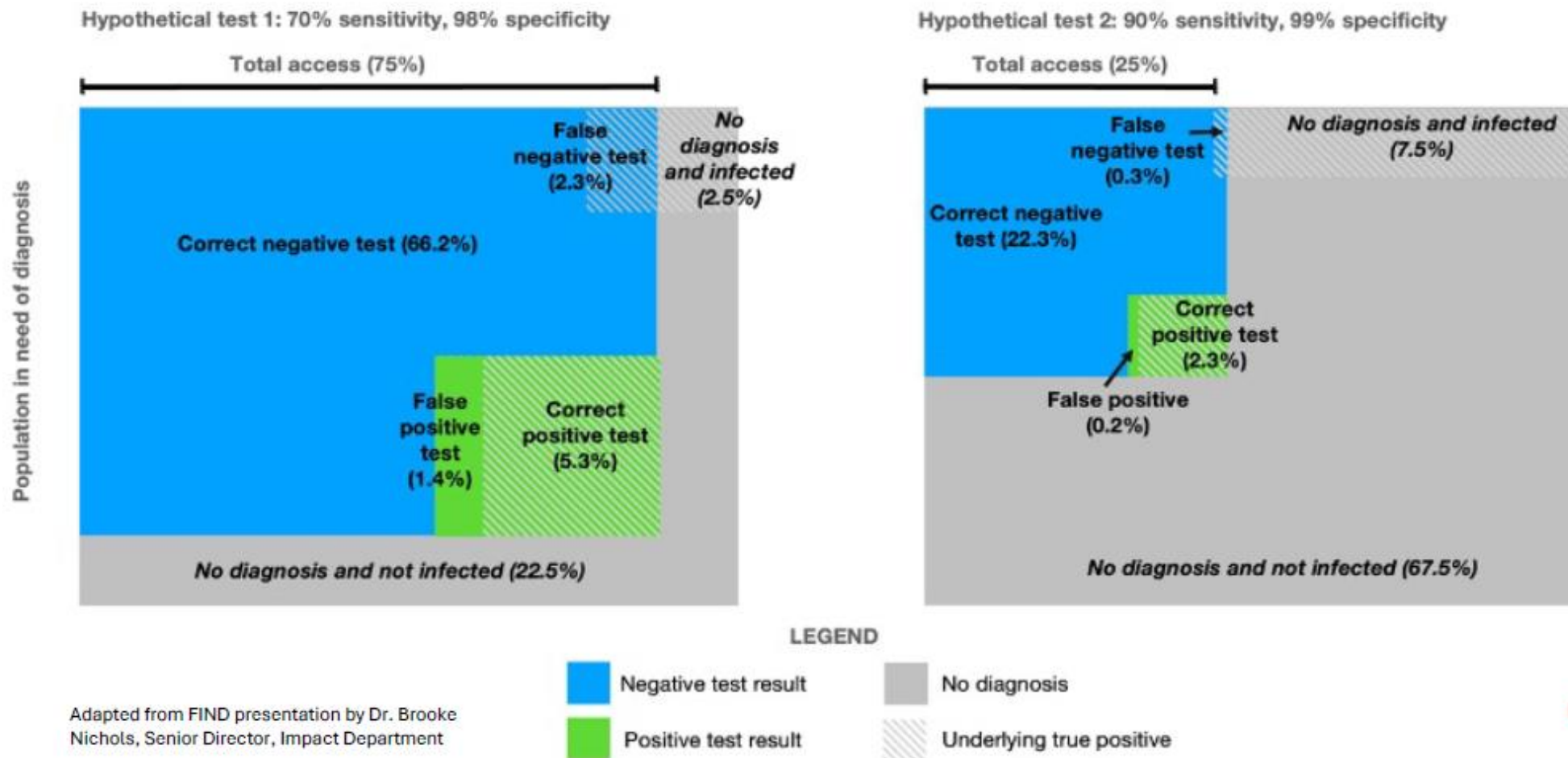
Interferon gamma release assay(IGRA)

1. QuantiFERON-TB GOLD (Cellestis Ltd, Australia)
2. T SPOT-TB (Oxford Immunodec, oxford, UK)
3. Immu-check TB Platinum (Immunoshop India Pvt Ltd)
4. TB-IGRA (Beijing Wantai Biological Pharmacy Enterprise Co. Ltd China)
5. ASACIR TB (Haikou VTI Biological Institute (China)



TB Diagnosis and the health system.

It's not just about accuracy, but also about accessible sample type

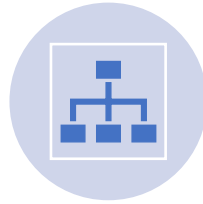


Adapted from FIND presentation by Dr. Brooke Nichols, Senior Director, Impact Department

TB Diagnosis and the health system.



Costs



Systems Readiness/
Useability



Strategic Fit - POC



Reliability



Coverage



Multiple functionalities

Conclusion

TB diagnostics limitation is the biggest gap in the care cascade gap

Thousands of innovations, A few reaching approval by WHO

Molecular testing , by far the most accurate

Effective TB diagnosis is beyond the lab science.

THANK YOU

